

Ben-Gurion University of the Negev

[Faculty or school]

[Department]

Correct Expected-Return Policy Inference in Stochastic Markov Decision Processes

Thesis draft

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Author: Eden Havri

Advisor: Ronen Brafman

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Abstract

This manuscript studies inference over deterministic policies while preserving the ordinary expected-return criterion of a stochastic Markov decision process. The scientific target is a Gibbs distribution whose log density is proportional to a policy's expected return, not to an exponentiated realized trajectory return. That distinction determines which Monte Carlo estimators are valid. The proposed approach combines a direct policy-space variational objective with a replicated particle method that averages each policy's return over common simulator tapes before exponentiation. Both maintain one sampled policy across an episode. The empirical program measures control performance, computational cost, and fidelity to the intended policy distribution.

Contents

Chapter 1

Introduction

Planning in a stochastic environment requires two conceptually separate operations. First, a policy is evaluated by averaging its return over the environment’s transition randomness. Second, a decision rule selects among policies. Conflating these operations changes the objective: exponentiating a single realized return before averaging is generally not equivalent to exponentiating the expected return.

This manuscript develops a policy-inference formulation in which the target is defined before an approximation algorithm is chosen. The target assigns mass to complete deterministic policies according to expected return. Its modes are therefore expected-return maximizers, while a finite inverse temperature retains a distribution over alternatives. The corrected inference method uses rollouts only through estimators that are valid for that target.

The study has four goals:

1. specify the expected-return policy distribution and its assumptions;
2. construct exact diagnostic oracles that detect target substitution;
3. derive direct and particle estimators with a consistent execution rule; and
4. compare target fidelity, control performance, and computational cost under matched experimental conditions.

Agreement with exact enumeration establishes target correctness before any large-scale performance comparison is attempted.

Chapter 2

Problem Setting and Notation

2.1 Finite-horizon stochastic control

Let an episodic Markov decision process be $\mathcal{M} = (\mathcal{S}, \mathcal{A}, T, R, \rho_0, H)$, where $T(\cdot \mid s, a)$ is a transition kernel, $R(s, a, s')$ is a bounded reward, ρ_0 is the initial-state distribution, and H is the rollout horizon. Time-to-go is included in the state whenever a stationary representation would otherwise exclude an optimal finite-horizon policy.

For a deterministic policy $\pi \in \Pi$, a trajectory $\tau = (s_1, a_1, \dots, s_{H+1})$ is generated by

$$s_1 \sim \rho_0, \quad a_t = \pi(s_t), \quad s_{t+1} \sim T(\cdot \mid s_t, a_t).$$

Its truncated return and the policy value are

$$G_H(\tau) = \sum_{t=1}^H R(s_t, a_t, s_{t+1}), \quad J_H(\pi) = \mathbb{E}_{\tau \sim p(\cdot \mid \pi)}[G_H(\tau)]. \quad (2.1)$$

2.2 Policy-level randomness

A distribution over deterministic policies is different from a Markov stochastic policy. Sampling one deterministic policy at the start of an episode creates correlations across states and revisits. Sampling a new action from a statewise marginal on every visit removes those correlations. Both execution rules can be studied, but they must not be treated as interchangeable.

Assumption 2.2.1 (Reference measure). *The deterministic policy class has a specified probability measure μ . For a finite tabular class, μ may be uniform. For a structured or continuous class, the choice of μ is part of the model and must be reported.*

Chapter 3

Expected-Return Policy Target

For inverse temperature $\beta > 0$, define the target measure

$$P_\beta(d\pi) = \frac{\exp(\beta J_H(\pi))}{Z_\beta} \mu(d\pi), \quad Z_\beta = \int_{\Pi} \exp(\beta J_H(\pi)) \mu(d\pi). \quad (3.1)$$

Bounded rewards and a finite reference measure make Z_β finite. The parameter β carries the inverse units of reward, so a change of reward units must be accompanied by the inverse change in β .

Proposition 3.0.1 (Mode preservation). *Every mode of P_β with respect to a uniform finite reference measure maximizes J_H , and every expected-return maximizer is a mode.*

Proof. The exponential function is strictly increasing, and the normalizing constant does not depend on π . $\square$

The order of expectation and exponentiation is essential. In general,

$$\exp(\beta \mathbb{E}[G_H]) \neq \mathbb{E}[\exp(\beta G_H)]. \quad (3.2)$$

The right-hand side defines an entropic-risk quantity and places additional weight on return variability. It is not an estimator identity for the target in Equation (??).

The target also has the variational characterization

$$\log Z_\beta = \sup_Q \{ \beta \mathbb{E}_{\pi \sim Q}[J_H(\pi)] - \text{KL}(Q \parallel \mu) \}. \quad (3.3)$$

Thus the regularization acts on a distribution over complete policies. It is not the same as adding independent action entropy at every visited state.

Chapter 4

Exact Validation Framework

The method is validated first on finite MDPs for which every deterministic policy can be enumerated. For each policy π , finite-horizon dynamic programming computes $J_H(\pi)$. Log-sum-exp normalization then gives the exact partition function, policy probabilities, and action marginals in Equation (??).

The smallest diagnostic contains two policies with equal expected return but different return variance. The target in Equation (??) assigns them equal probability under a uniform reference measure. A method that instead uses the right-hand side of Equation (??) prefers the variable policy. This is a deterministic pass-or-fail test of the order of operations.

Further diagnostics test state revisitation, absorbing states, policy memoization, particle ancestry, common random numbers, and exact gradients. No cluster-scale experiment is run until:

- transition kernels and rewards pass deterministic unit tests;
- direct-objective gradients agree with exhaustive differentiation;
- particle marginals agree with the exact finite-sample target; and
- increasing the number of simulator tapes reduces expected target error.

Chapter 5

Corrected Inference Method

5.1 Direct stochastic variational objective

Let q_θ be a tractable distribution over complete deterministic policies. The corrected method maximizes the evidence lower bound implied by Equation (??):

$$\mathcal{L}(\theta) = \beta \mathbb{E}_{\pi \sim q_\theta} [J_H(\pi)] - \text{KL}(q_\theta \| \mu). \quad (5.1)$$

Sampling $\pi \sim q_\theta$ and then $\tau \sim p(\cdot | \pi)$ gives an unbiased estimator of the return term because Equation (??) is linear in J_H .

For a score-function estimator, a valid single-sample gradient contribution is

$$(\beta G_H(\tau) + \log \mu(\pi) - \log q_\theta(\pi) - b) \nabla_\theta \log q_\theta(\pi), \quad (5.2)$$

where b is an action-independent control variate. Multiple independent rollouts of the same policy may be averaged before applying Equation (??); this reduces variance without changing the target.

5.2 Policy representation and execution

For a finite tabular policy, actions may be sampled lazily and memoized on the first visit to each state. The log proposal probability contains each sampled state assignment exactly once. Particle resampling, when used as an auxiliary search mechanism, must copy the state, policy memo, visit counters, and any other ancestry-dependent state together.

The primary execution rule samples one policy-level object at the beginning of an episode and preserves it until termination. A separately named *marginal-action*

ablation samples from estimated statewise marginals on every visit. Comparing these two rules measures the value of policy-level correlation instead of assuming it away.

5.3 Role of common random numbers

Shared random streams may compare candidate policies under matched transition noise. They are treated as a variance-reduction design, not as part of the probabilistic target. A correct estimator should converge to the same result under shared and independent streams; finite-sample efficiency may differ.

5.4 Replicated sample-average policy SMC

For a fixed bank of K simulator tapes $\Xi_K = (\xi_1, \dots, \xi_K)$, define

$$\hat{J}_K(\pi; \Xi_K) = \frac{1}{K} \sum_{k=1}^K G_H(\pi, \xi_k). \quad (5.3)$$

The conditional sample-average target is

$$P_{\beta, K}(d\pi \mid \Xi_K) \propto \exp(\beta \hat{J}_K(\pi; \Xi_K)) \mu(d\pi). \quad (5.4)$$

5.5 Learned full-support proposal

Let m denote the assignments already stored in a particle’s policy memo, and let $\mathcal{A}_\mu(s, m)$ be the prior-supported legal actions. A stochastic policy $q_{\text{PPO}, \phi}$, trained with PPO, is used as a guide rather than as the inference target. With $u_\mu(\cdot \mid s, m)$ uniform on $\mathcal{A}_\mu(s, m)$, the actual proposal is

$$q_{\epsilon, \phi}(a \mid s, m) = (1 - \epsilon) q_{\text{PPO}, \phi}(a \mid s, m) + \epsilon u_\mu(a \mid s, m), \quad 0 < \epsilon \leq 1. \quad (5.5)$$

Both components use the same legal-action mask. Hence every action in the target support has proposal probability at least $\epsilon/|\mathcal{A}_\mu(s, m)|$. The mixture probability in Equation (??), rather than the unmixed PPO probability, is the denominator in every importance ratio.

Each particle contains one policy memo and K environment replicas. Let U_t be the deterministically ordered set of distinct unassigned states first exposed by those replicas at simulation step t . One action is drawn and memoized for each $s \in U_t$.

The exact incremental log-weight is

$$\Delta \log w_t = \frac{\beta}{K} \sum_{k=1}^K r_{t,k} + \sum_{s \in U_t} [\log \mu(a_s \mid s, m_{<s}) - \log q_{\epsilon, \phi}(a_s \mid s, m_{<s})]. \quad (5.6)$$

Thus the mean reward is formed before exponentiation, and each prior/proposal correction is applied once per newly instantiated policy assignment, not once per replica or visit. Revisits reuse the memoized action. Resampling copies all replica states, the complete policy memo, and proposal context. No transition ratio is required because the conditional target and proposal use the same simulator dynamics.

The guide is frozen within each SMC sweep, and the parameters, action mask, and value of ϵ used to generate every proposal are recorded. Proposal adaptation between sweeps, if studied, uses a separate frozen snapshot for each sweep.

For finite K , Equation (??) is an explicit approximation, not an unbiased representation of Equation (??). Under bounded returns and a finite policy class it converges uniformly as K increases. Experiments therefore report convergence in K separately from convergence in the number of policy particles.

5.6 Proposal accounting and isolation

Training $q_{\text{PPO}, \phi}$ is part of the guided method. Matched-budget comparisons count all simulator interactions, optimizer updates, wall-clock time, accelerator time, and peak memory used to train the proposal as well as the later SMC cost. Any amortized accounting is reported only as a separately specified multi-instance setting.

The required isolation study includes uniform-proposal SMC ($\epsilon = 1$), corrected guided SMC using Equations (??)–(??), and a proposal-alone condition. The proposal-alone condition samples the same lazy, memoized deterministic policy from $q_{\epsilon, \phi}$ but performs no importance weighting or resampling. It therefore isolates the contribution of SMC from that of the learned guide while preserving policy-commitment semantics. These conditions define the evaluation protocol; they do not state an empirical outcome.

Chapter 6

Experimental Design

The experimental program begins with exact target-fidelity gates, proceeds to small end-to-end learning tests, and only then compares methods under matched environments and compute budgets.

6.1 Target-fidelity tests

Small finite MDPs permit enumeration of deterministic policies and direct calculation of Equation (??). They provide the primary correctness tests:

- total-variation or Kullback–Leibler error to the exact policy target;
- recovery of target modes and statewise marginals;
- sensitivity to transition variance at fixed expected return;
- invariance to random-number coupling in the large-sample limit; and
- agreement between memoized-policy execution and the predicted trajectory distribution.

6.2 Control and scaling tests

Larger domains measure expected return, success probability, lower-tail return, simulator calls, wall-clock time, and peak memory. Difficulty should vary along interpretable axes: state-space size, obstacle density, horizon, transition stochasticity, delayed reward, and number of policy modes.

Every comparison uses the same training seeds, evaluation seeds, rollout horizon, and termination convention. Hyperparameter search budgets are matched and reported separately from final evaluation. Reward scaling is paired with the inverse scaling of β whenever the intended target is held fixed.

Table 6.1: Minimum result matrix. Each cell should report central tendency, uncertainty, and compute cost over independent runs.

Question	Required baseline	Primary metric
Implementation validity	exact oracle	deterministic agreement
Target correctness	exact enumeration	distribution error
Control quality	planning baselines	expected and tail return
Scalability	matched compute	time and memory
Execution semantics	memoized vs marginal	trajectory divergence

Chapter 7

Discussion and Limitations

The corrected formulation preserves expected-return ordering at the level of target modes. A finite-temperature draw is nevertheless not an expected-return-optimal policy in the classical sense; it is a randomized selection rule whose behavior depends on β and the reference measure. Those modeling choices must be stated rather than described as data-driven epistemic uncertainty.

The direct objective in Equation (??) avoids a nonlinear transformation of a noisy log density, but it can still have high gradient variance and a restrictive variational family. A factorized statewise proposal cannot represent correlations between alternative policy modes. Structured global latent variables, autoregressive assignments, or retained weighted policy particles are possible extensions, provided their objectives continue to target Equation (??).

Finally, simulator access makes the method a form of planning with a generative model. It should not be described as model-free merely because transition probabilities are never enumerated.

Chapter 8

Conclusion

Expected-return policy inference is well defined once the target measure, reference measure, inverse temperature, and execution semantics are explicit. The central correction is methodological: an unbiased estimator of expected return may be used linearly in the policy-space variational objective, but it cannot be exponentiated inside a multi-sample log average without changing the quantity being optimized. Exact finite-policy gates and convergence-controlled sample averages make that claim directly testable and give the empirical results a clear interpretation.

Core Derivations

Gibbs variational identity. Let P_β be the target in Equation (??). For any $Q \ll \mu$, expand

$$\text{KL}(Q \| P_\beta) = \mathbb{E}_Q \left[\log \frac{dQ}{d\mu} - \beta J_H(\pi) + \log Z_\beta \right] \quad (1)$$

$$= \log Z_\beta - (\beta \mathbb{E}_Q[J_H(\pi)] - \text{KL}(Q \| \mu)). \quad (2)$$

Non-negativity of the left-hand side yields Equation (??), with equality at $Q = P_\beta$.

Why a rollout is valid in the direct objective. For $\pi \sim q_\theta$ and $\tau \sim p(\cdot | \pi)$, iterated expectation gives

$$\mathbb{E}_{\pi, \tau}[G_H(\tau)] = \mathbb{E}_{\pi \sim q_\theta}[J_H(\pi)].$$

No exponential or logarithmic transformation is applied between the inner and outer expectations.